

NORTHERN UNIVERSITY BANGLADESH

Department of Computer Science & Engineering

Course: Database Management System (DBMS)

PROJECT PROPOSAL

University Management System

Technology Stack: SQL/MySQL (Database), HTML, Bootstrap, CSS (Frontend/UI), PHP (Backend)

1. Project Overview

The University Management System is a web-based database application designed to efficiently manage university academic and administrative activities. The system provides a centralized platform to store, organize, update, and retrieve important university information including students, faculty members, departments, programs, courses, registrations, payments, admit cards, and final results.

This project highlights the importance of Database Management Systems by ensuring structured data storage, data consistency, security, and efficient information management for university operations.

2. System Features & Modules

Student Management: Manages student information, academic records, and related details.

Faculty Management: Maintains faculty information and department-related data.

Department & Program Management: Handles departments, programs, academic sessions, and semesters.

Course Management: Manages available courses and course information.

Course Registration Module: Handles student course enrollment and registration details.

Payment Management: Maintains payment structures, student payments, and financial records.

Admit Card Management: Generates and manages student admit card information.

Result Management: Stores and manages final academic results.

Employee & User Management: Handles employees, authentication, and user access control.

3. Database Tables Description

Table Name	Purpose
student	Stores student personal and academic information.
faculty	Stores faculty member information.
programs	Stores university programs information.
department	Stores department details.
session	Stores academic session information.

Table Name	Purpose
semester	Stores semester details.
admit_card	Stores student admit card information.
course	Stores course details.
course_registration	Stores student course registration information.
course_registration_details	Stores detailed course enrollment data.
payment_structure	Stores payment rules and fee structure.
payment	Stores student payment records.
final_result	Stores final academic results.
employee	Stores employee information.
user	Stores system login and user management data.

4. Future Improvements

The system can be enhanced in the future with online payment gateway integration, mobile application support, attendance management, notification services, and advanced reporting and analytics features.

5. Conclusion

The University Management System project aims to provide an efficient, secure, and organized solution for managing university activities digitally. By applying database management concepts, this system will improve data accuracy, accessibility, and operational efficiency while reducing manual work. The expected outcome is a reliable platform that supports academic and administrative management in a modern university environment.